Basic Computer Terms

Computer

An electronic machine that takes input, processes it, and gives output.
It works very fast and accurately.
Example: Laptop, desktop.

Hardware

Physical parts of a computer you can touch.
Includes keyboard, mouse, monitor, CPU.
Without hardware, a computer cannot exist.

Software

Programs that tell hardware what to do.
It runs inside the computer.
Example: MS Word, browser.

Data

Raw facts or unprocessed information.
It has no meaning by itself.
Example: Numbers, names.

Information

Processed and meaningful data.
It helps in decision making.
Example: Report, result.

Input Device

Used to enter data into a computer.
Example: Keyboard, mouse, scanner.
Helps user interact with system.

Output Device

Shows results from the computer.
Example: Monitor, printer, speaker.
Displays processed data.

CPU (Central Processing Unit)

The brain of the computer.
Processes all instructions and data.
Controls all operations.

Memory

Stores data and instructions temporarily.
Used while working on tasks.
Example: RAM.

Storage

Stores data permanently.
Keeps files even after shutdown.
Example: HDD, SSD.

💾 Memory & Storage

RAM (Random Access Memory)

Temporary memory used while working.
Very fast but data is lost after shutdown.
Helps run programs smoothly.

ROM (Read Only Memory)

Permanent memory.
Stores important startup instructions.
Data cannot be easily changed.

Cache Memory

Very fast memory near CPU.
Stores frequently used data.
Improves performance.

HDD (Hard Disk Drive)

Uses magnetic storage.
Slower compared to SSD.
Used for long-term storage.

SSD (Solid State Drive)

Faster than HDD.
No moving parts, more reliable.
Used in modern computers.

Byte / KB / MB / GB / TB

Units to measure data size.
1 KB = 1024 bytes, 1 MB = 1024 KB.
Used to measure storage capacity.

🖥️ Operating System & Software

Operating System (OS)

Main software that controls the computer.
Manages hardware and software.
Example: Windows, Linux.

Windows / Linux / macOS

Popular operating systems.
Each has different features.
Used on computers worldwide.

Application Software
Programs used for specific tasks.
Example: MS Word, browser.
Helps users perform work.
System Software
Controls system operations.
Includes OS and drivers.
Works in the background.
Driver
Software that connects hardware with OS.
Helps devices work properly.
Example: Printer driver.
Booting
Process of starting a computer.
Loads operating system.
Happens when you turn on PC.
Multitasking
Running multiple programs at once.
Example: Music + browser together.
Saves time and improves efficiency.

🌐 Internet & Networking

Internet
Global network connecting computers.
Used to share information.
Example: Websites, emails.
Website
Collection of web pages.
Accessible through internet.
Example: Google, YouTube.
Web Browser
Software to open websites.
Example: Chrome, Firefox.
Helps access internet.
URL
Address of a website.
Typed in browser.
Example: www.google.com.
HTTP / HTTPS
Rules for web communication.
HTTPS is secure (encrypted).
Used while browsing.
IP Address
Unique identity of a device.
Helps identify systems on network.
Example: 192.168.1.1.
Wi-Fi
Wireless internet connection.
No cables required.
Used in homes and offices.
LAN / WAN
Types of networks.
LAN = small area, WAN = large area.
Example: Office vs Internet.
Cloud Computing
Storing data online.
Access from anywhere.
Example: Google Drive.

🔐 Cyber Security Basics

Password
Secret code for access.
Protects accounts and data.
Should be strong and unique.
Authentication
Verifying identity.
Example: Password, OTP.
Ensures security.
Firewall
Protects network from threats.
Blocks unauthorized access.
Acts like a security guard.
Antivirus
Protects from viruses.
Scans and removes threats.
Example: Quick Heal.
Malware
Harmful software.
Damages system or steals data.
Includes virus, worm.

Virus / Trojan / Worm
Types of malware.
Spread in different ways.
Cause damage to system.

Phishing
Fake attempt to steal data.
Looks like real messages.
Example: Fake bank emails.

Encryption
Converts data into secret form.
Protects sensitive data.
Used in secure communication.

🧑‍💻 Programming & IT Concepts

Programming Language
Language to write code.
Example: Python, Java.
Used to build software.

Algorithm
Step-by-step solution.
Helps solve problems logically.
Used before coding.

Flowchart
Diagram of program logic.
Uses symbols and arrows.
Makes understanding easy.

Compiler
Converts full code into machine language.
Runs whole program at once.
Example: C compiler.

Interpreter
Runs code line by line.
Easier for debugging.
Example: Python.

Source Code
Human-written program code.
Written using languages.
Converted into machine code.

Bug
Error in program.
Causes wrong output.
Needs fixing.

Debugging
Process of fixing bugs.
Improves program quality.
Done by developers.

API
Connects different software.
Allows communication between apps.
Example: Payment gateway API.

Database
Stores organized data.
Easy to manage and access.
Example: MySQL.

🤖 Modern & Emerging Terms

Artificial Intelligence (AI)
Machines that act like humans.
Can think and decide.
Example: Chatbots.

Machine Learning (ML)
Learning from data automatically.
Improves with experience.
Used in predictions.

Data Science
Analyzing data for insights.
Uses statistics and tools.
Helps decision making.

Big Data
Very large data sets.
Difficult to process normally.
Used in analytics.

Blockchain
Secure digital record system.
Data stored in blocks.
Used in cryptocurrency.

IoT (Internet of Things)
Smart connected devices.
Communicate via internet.
Example: Smart TV.

Automation
Work done without human effort.
Saves time and cost.
Example: Robots.
Virtual Reality (VR)
Virtual world experience.
Needs special headset.
Used in gaming.
Augmented Reality (AR)
Adds digital objects to real world.
Example: Filters in apps.
Enhances real environment.

📊 Exam & Career Focus
Digital Literacy
Ability to use digital tools.
Important in modern life.
Includes basic computer skills.
E-mail
Electronic message system.
Used for communication.
Example: Gmail.
Spreadsheet
Used for calculations and data.
Example: Excel.
Used in business.
Presentation Software
Used to create slides.
Example: PowerPoint.
Used in meetings.
Backup
Copy of important data.
Prevents data loss.
Stored safely.
Update
Improves software.
Fixes bugs and adds features.
Important for security.
Open Source
Free software with editable code.
Anyone can modify it.
Example: Linux.
Version Control
Tracks code changes.
Helps manage development.
Used in teams.
Git
Popular version control tool.
Tracks and manages code.
Used by developers worldwide.

Basic Computer Terms

1. What is a computer?
   A) Mechanical device
   B) Electronic machine that processes data
   C) Only storage device
   D) Output device
2. Which of the following is hardware?
   A) MS Word
   B) Windows
   C) Keyboard
   D) Chrome
3. Software is:
   A) Physical part
   B) Set of programs
   C) Input device
   D) Storage unit
4. Data means:
   A) Processed result
   B) Raw facts
   C) Final output
   D) Software
5. Information is:
   A) Raw data
   B) Meaningful data
   C) Hardware
   D) Memory
6. Which is an input device?
   A) Monitor

B) Printer
C) Keyboard
D) Speaker

7. Which is an output device?
   A) Mouse
   B) Keyboard
   C) Monitor
   D) Scanner
8. CPU is known as:
   A) Memory
   B) Brain of computer
   C) Storage
   D) Input unit

💾 Memory & Storage

9. RAM is:
   A) Permanent
   B) Temporary memory
   C) Slow memory
   D) Storage device
10. ROM is:
   A) Temporary
   B) Permanent memory
   C) Cache
   D) Input device
11. Cache memory is:
   A) Slow
   B) External
   C) Very fast
   D) Not used
12. Which is faster?
   A) HDD
   B) SSD
   C) CD
   D) DVD
13. 1 KB = ?
   A) 100 bytes
   B) 512 bytes
   C) 1024 bytes
   D) 2048 bytes

💻 Operating System & Software

14. OS stands for:
   A) Open Software
   B) Operating System
   C) Output System
   D) Office System
15. Which is an OS?
   A) MS Word
   B) Windows
   C) Chrome
   D) Excel
16. Application software example:
   A) Linux
   B) Driver
   C) MS Word
   D) BIOS
17. Driver is used to:
   A) Play music
   B) Connect hardware with OS
   C) Store data
   D) Run browser
18. Booting means:
   A) Shutdown
   B) Restart
   C) Starting computer
   D) Formatting
19. Multitasking means:
   A) One task
   B) No task
   C) Multiple programs at once
   D) Only gaming

🌐 Internet & Networking

20. Internet is:
   A) Local system
   B) Global network
   C) Hardware
   D) Software
21. A website is:
   A) Software
   B) Collection of web pages

C) Input device
D) OS

22. Which is a web browser?
A) Windows
B) Chrome
C) Linux
D) CPU

23. URL is:
A) Device
B) Address of website
C) Software
D) Memory

24. HTTPS is:
A) Unsafe
B) Secure
C) Slow
D) Offline

25. IP Address is:
A) Password
B) Device identity
C) Software
D) Virus

🔐 Cyber Security

26. Password is used for:
A) Playing games
B) Security
C) Storage
D) Output

27. Firewall is:
A) Virus
B) Security barrier
C) Input device
D) Software

28. Antivirus is used to:
A) Create virus
B) Delete files
C) Protect system
D) Increase speed

29. Malware is:
A) Useful software
B) Harmful software
C) Hardware
D) Driver

30. Phishing is:
A) Fishing
B) Fake data stealing method
C) Storage
D) Backup

🧑‍💻 Programming & IT

31. Python is:
A) Hardware
B) Programming language
C) OS
D) Browser

32. Algorithm is:
A) Code
B) Step-by-step solution
C) Error
D) Output

33. Flowchart is:
A) Code
B) Diagram of logic
C) Data
D) Memory

34. Compiler:
A) Runs line by line
B) Converts full code
C) Deletes code
D) Debugs

35. Interpreter:
A) Converts whole program
B) Runs line by line
C) Stores data
D) Output

36. Bug means:
A) Virus
B) Error in program
C) Hardware
D) OS

37. Debugging means:
    A) Writing code
    B) Fixing errors
    C) Deleting code
    D) Running code

🤖 Modern Concepts

38. AI means:
    A) Artificial Intelligence
    B) Automatic Input
    C) Advanced Internet
    D) Application Interface
39. ML is:
    A) Manual Learning
    B) Machine Learning
    C) Memory Load
    D) Main Logic
40. IoT means:
    A) Internet of Things
    B) Input of Tools
    C) Internal Output Tech
    D) None
41. Blockchain is:
    A) Storage
    B) Secure record system
    C) Virus
    D) Browser

📊 Exam Focus

42. Email is used for:
    A) Storage
    B) Communication
    C) Input
    D) Output
43. Spreadsheet example:
    A) Word
    B) Excel
    C) Chrome
    D) Linux
44. Backup means:
    A) Delete data
    B) Copy of data
    C) Format
    D) Update
45. Open Source means:
    A) Paid software
    B) Free editable software
    C) Virus
    D) Driver
46. Git is:
    A) Browser
    B) Version control tool
    C) OS
    D) Hardware

✅ Answer Key

1-B, 2-C, 3-B, 4-B, 5-B
6-C, 7-C, 8-B
9-B, 10-B, 11-C, 12-B, 13-C
14-B, 15-B, 16-C, 17-B, 18-C, 19-C
20-B, 21-B, 22-B, 23-B, 24-B, 25-B
26-B, 27-B, 28-C, 29-B, 30-B
31-B, 32-B, 33-B, 34-B, 35-B, 36-B, 37-B
38-A, 39-B, 40-A, 41-B
42-B, 43-B, 44-B, 45-B, 46-B